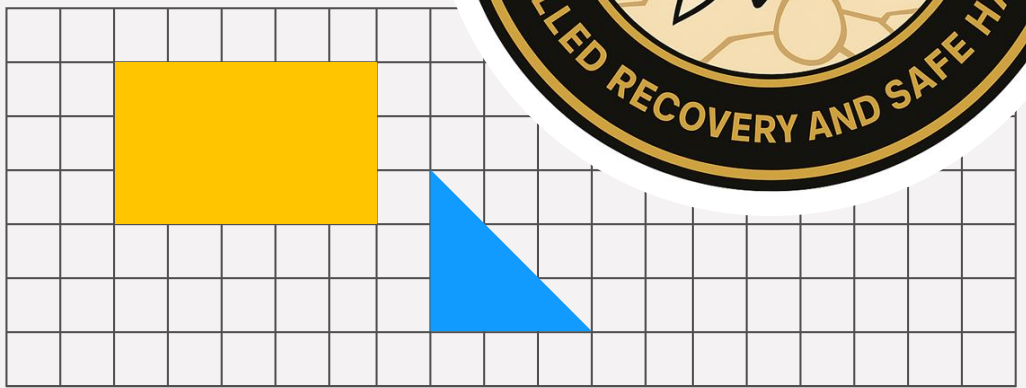
# CRASH TEAM TWO

The CRASH competition challenges teams to design an Level 1 class rocket that can safely deliver fragile payloads

Prepared by:
John “Yianni” Bray
Logan Hammer
Camila Martinez
Dylan Moga
Christian Warson

Faculty Advisor:
Adam Marquardt



# Overview



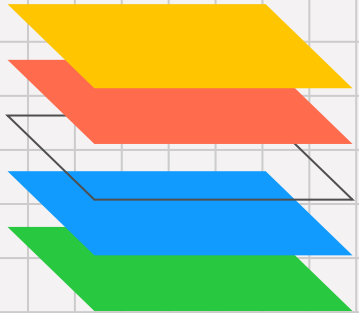

- CRASH is focused on payload handling
- Main challenge is the safe recovery of eggs
- Heavy focus on weight-critical design

## Presentation

- Competition Specific Requirements
- General System Architecture
- Detailed Subsystem Design Overviews
  - Requirements/Risks/Mitigation
- Simulation Analysis
- Gantt Chart and the Critical Path

# The Competition



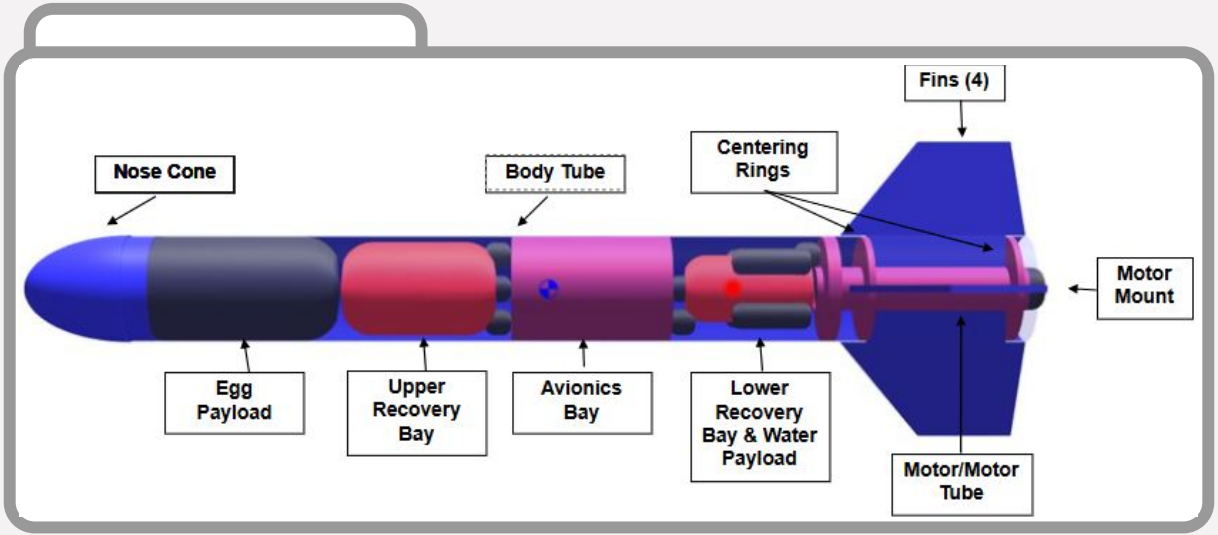


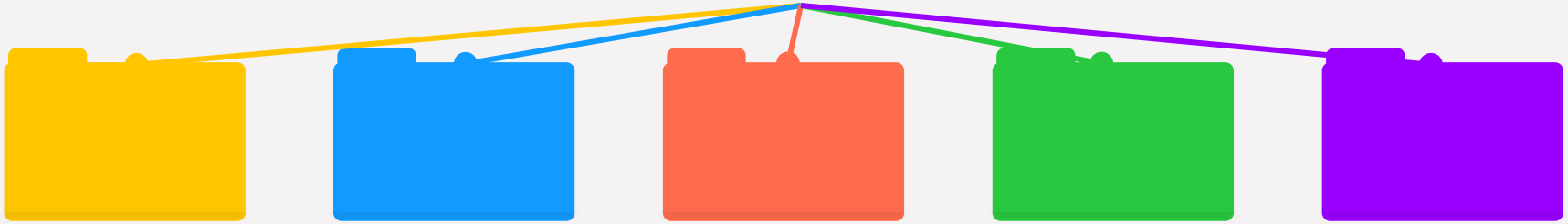
Fragile!!

- ★ Altitude between 1000 and 1500 feet
- ★ 800 feet or less from launch pad
- ★ Less than 25 ft/s during final descent
- ★ At least 5 ounces of water
- ★ At least 2 eggs
- ★ Recovered in flight worthy condition

The goal of the CRASH **(Controlled Recovery and Safe Handling)** is a rocket competition hosted by the University of Central Florida with the goal of introducing students to the engineering principles behind **launching, protecting, and safe recovery of fragile payloads.**

# Architecture



Egg Payload

Water Payload

Recovery & Propulsion

Aerostructures

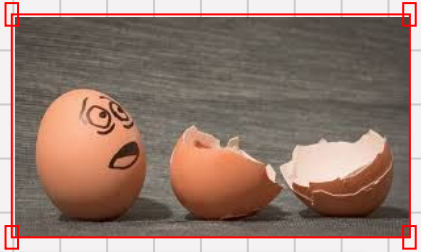
Avionics

# Egg Payload

## Requirements

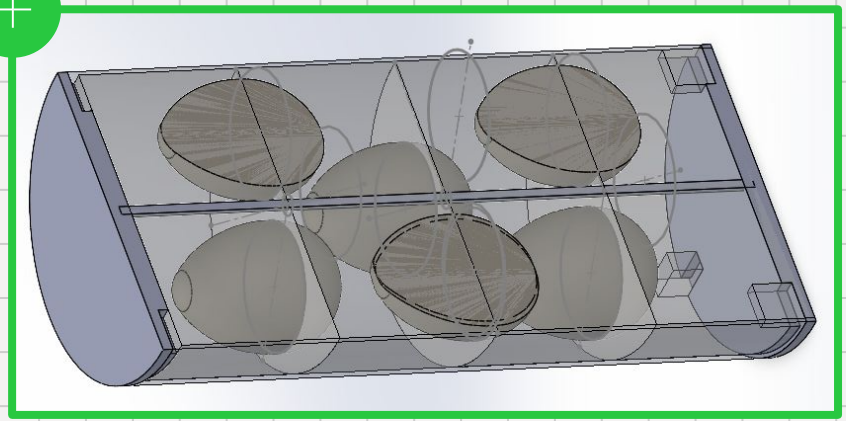
| |
|---|
| The egg protection system **shall** allow for eggs to be loaded and unloaded within 5 minutes. |
| The egg payload section **shall** weigh no more than 30 oz. |
| Egg payload location **shall** be consistent compared to pre/post flight. |
| The egg protection system **shall** reduce vibration intensity to less than or equal to 7.5 m/s². |
| The egg protection system **shall** limit impact forces to less than 250N. |
| Holding vessels for eggs **shall** be free of cracks, dents, leaks, holes, or bending. Minor cracks and dents may be permissible as long as the integrity of the structure is not compromised and the fragile cargo is unharmed. |

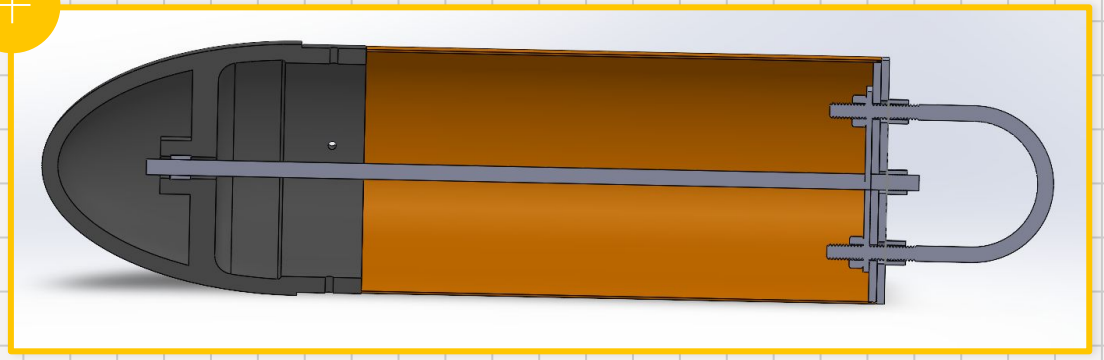
## Risks



| |
|---|
| Failure to dampen vibrations |
| Failure to dampen impact forces |
| Failure to properly secure eggs |
| Capsule stuck in the rocket |
| Capsule not opening |
| Egg pucks take too long to stack |
| Egg pucks take too long to unstack |
| Failure to secure the rod to the nosecone |
| Failure of the nose cone to withstand the snatch force |
| Failure for bulkheads to withstand the snatch force |
| Not enough Foam for the final design |

# Egg Payload Design and Modeling





## Internals

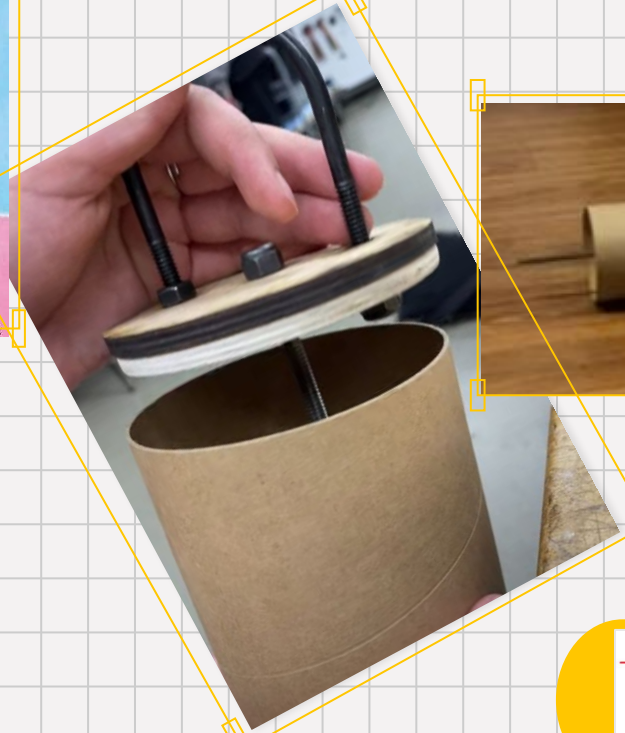
- Foam
- Aluminum Rod
- Eggs

## Externals

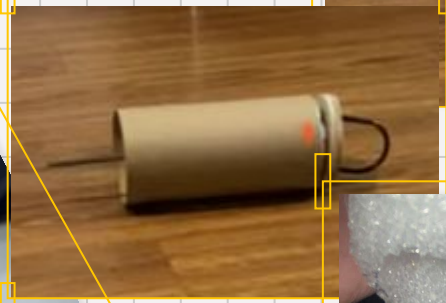
- Cardboard Coupler
- PLA Nose Cone
- U Bolt
- Nuts
- Wood Bulkheads

# Egg Payload Prototyping and Testing









## Test?

For testing we go back to competition roots and drop the payload from a significant height to reach a velocity of 24 m/s

The height is 9ft

It passes the test if all eggs survive!

**Free Fall Equations**

$$V_F = V_0 + gt$$

$$d = V_0 t + gt$$

# Water Payload

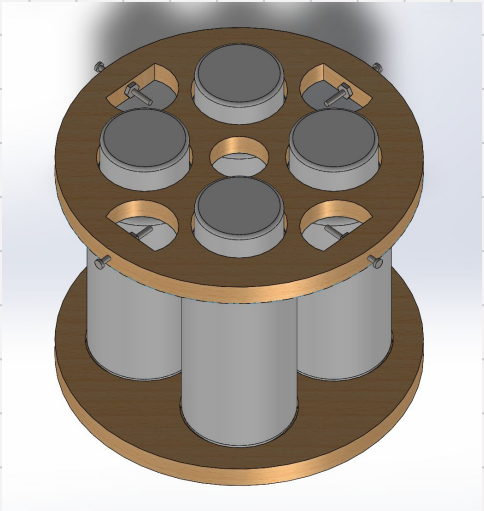
## Requirements

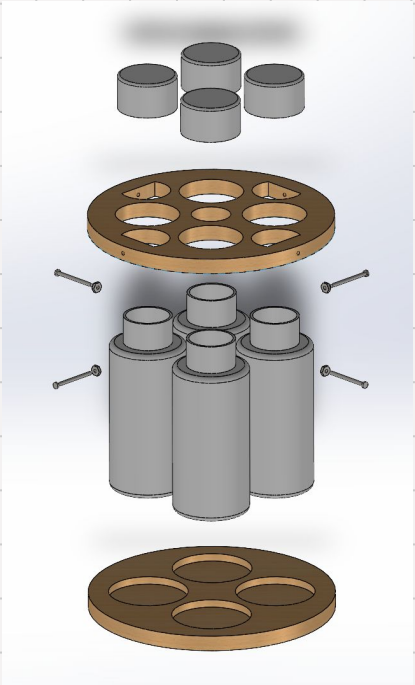
| |
|---|
| Target volume for water **shall** be no less than 5 fl oz. |
| Water payload section **shall** weigh no more than 7 oz. |
| Water payload location **shall** be consistent compared to pre/post flight. |
| Holding vessels for water **shall** be free of cracks, dents, leaks, holes, or bending. |
| Water vessels **shall** be able to be loaded/unloaded in at most 5 minutes |
| Water vessel supports **shall** be able to withstand 300N minimum |

## Risks

| |
|---|
| Leakage resulting in disqualification |
| Comprised structural integrity of water vessels |
| Thermal damage to water vessels from the motor burn |
| Vessel seals bursting under high intensity loading periods |
| Vessel seals failing due to improper tightening |
| Vessels becoming dislodged from supporting structure |
| Improper fitting of the water vessels due to manufacturing |
| Supporting ring failure due to improper bolt fastening |

# Water Payload





## Mitigation

- Materials Selection

## Components

- HDPE Vessels & Caps
- Steel Nuts & Bolts
- Birch Ring Supports

## Manufacturing

- COTS Products
- Custom Laser Cut Supports

# Water Payload



## Prototyping & Testing

- ❑ Partial Prototypes
- ❑ Verification & Validation
- ❑ Vessel Drop (Burst) Testing
- ❑ Support Stress Testing
- ❑ Water Deployment Method Testing

# Recovery & Propulsion

## Requirements

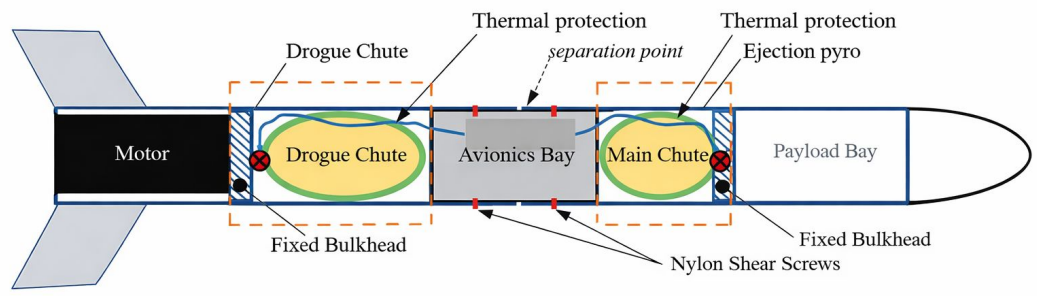
| |
|---|
| Launch vehicle apogee **shall** exceed 999 ft, while not exceeding 1500 ft. |
| Launch vehicle **shall** land within 800 ft of the launch pad. |
| Descent rate **shall** not exceed 25 ft/s during the last 5 seconds of flight. |
| Motor **shall** not be in an impulse class higher than H. (320 N*s max) |
| Parachutes **shall** be fully intact and untangled, free of any holes, shredding, and any damage to the suspension lines. |
| Deployment of recovery systems **shall** result in no damage to the airframe such as zippering on the body tube. |

## Risks

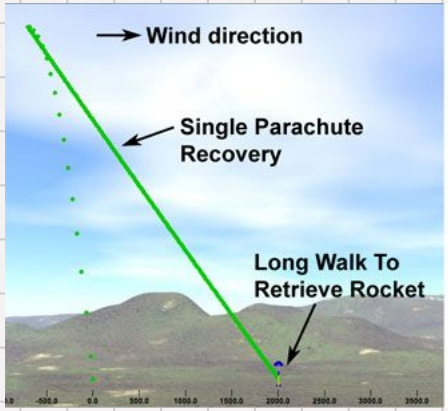
| |
|---|
| Damage to the parachute, suspension lines, or shock cord could lead to unsafe ground collision speeds. |
| Improper folding/packing of the parachutes and shock cords could lead to deployment failures. |
| Incorrectly sized parachutes will lead to unsafe ground collision speeds. |
| Early deployment of parachutes will lead to the rocket drifting beyond 800 feet. |
| Late deployment of parachutes will lead to unsafe ground collision speeds. |
| Motor ignition failure would lead to a failure to launch |
| Motor explosion would lead to a damaged rocket not recovered in flyable condition. |

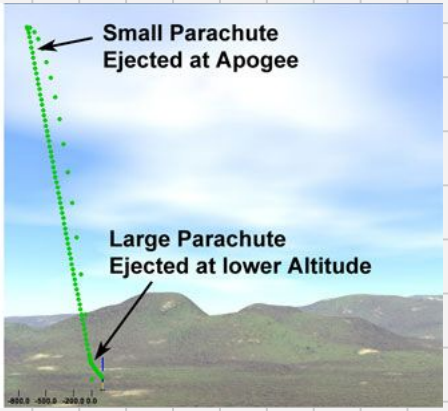
# Recovery & Propulsion

## Recovery Systems




| |
|---|
| We will be using a dual deployment system as our recovery method. |
| A smaller parachute deployed at high altitude to burn off speed without adding much distance. |
| A larger parachute deployed at low altitude to burn off remaining speed with the consequence of adding distance. |
| The small parachute will be 24” in diameter and the large parachute will be 78” in diameter. |
| Kevlar shock cords will connect the parachutes to the body of the rocket. |

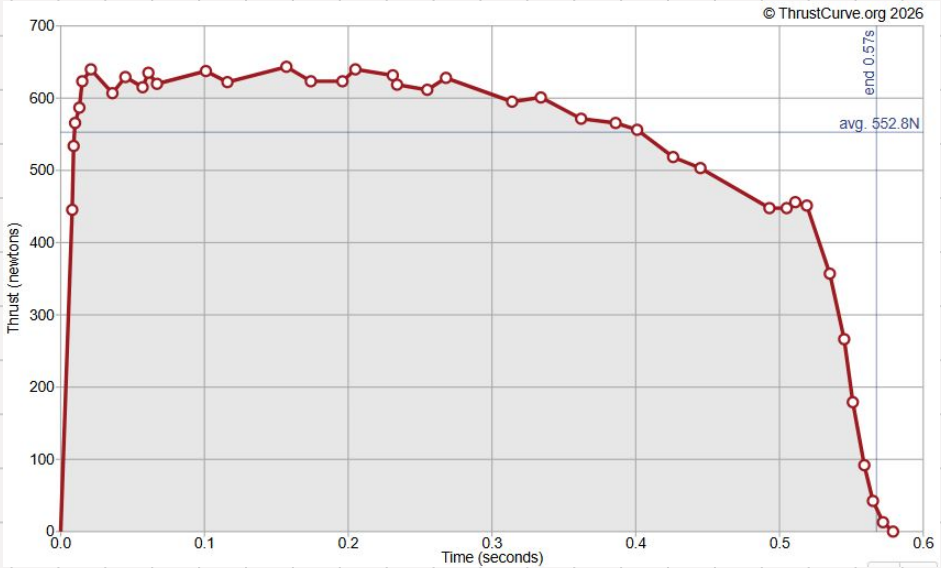


# Recovery & Propulsion

## Motor Selection

| |
|---|
| We will be using an AeroTech H550ST-14A motor. |
| We have chosen to use a Disposable Motor System as it is simpler and much more cost efficient for the purposes of our rocket. |
| This motor was selected since it is the only commercially available H class motor with an impulse of 320 Newton seconds. |
| The only concern with our motor is the peak thrust of 667N, but we believe we can cushion the egg payload enough to prevent damage. |



# Aerostructures

## Level 1 Requirements

| |
|---|
| Nose cone **shall** be fully intact pre-flight, free of any material loss, denting, and cracking. |
| Connection to the body tube **shall** be fixed and rooted under pressure. |
| The body tube **shall** be fully intact, free of any cracks, bubbling, or crushing. |
| Fins **shall** be fully intact pre & post flight with a strong connection to the body tube and experience no bending, warping, extensive crack propagation, or major chips. |
| Launch vehicle stability **shall** fall between the range of 1.5 - 2.0. Anything outside of this range may lead to failure. |

## Risks

| |
|---|
| Nose cone failure could prevent reuse or reduce stability |
| Body tube connection failure could cause instability; rocket crashes |
| Body tube defects could prevent reuse of the vehicle & reduce stability |
| Fin deformation could affect flight performance & stability |
| Stability failure is a major safety concern & could lead to hazardous conditions |
| Interior structure component failure could deem the rocket unflightworthy |

# Aerostructures

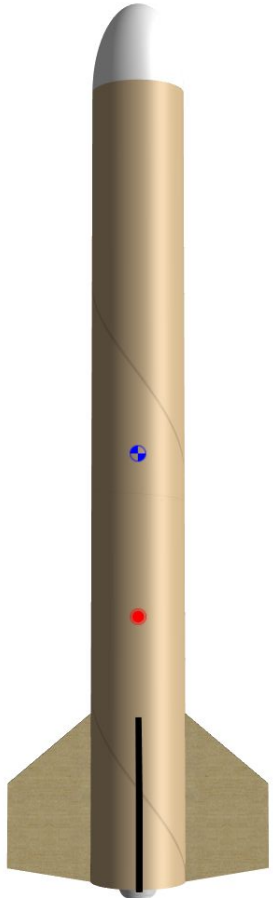
| Risks & Failure Modes | Mitigation Strategies |
|---|---|
| Body tube cracks | Handling the rocket carefully and controlling descent velocity |
| Body tube disconnects early | Ensure proper connection |
| Rail pins tear through body | Ensure proper connection and screw is not fretting |
| Nose cone buckles during flight | Ensure material is stable & has a strong bond |
| Nose cone cracks | Handling the rocket carefully and controlling descent velocity |
| Fins break off before/during ignition | Handling the rocket carefully and controlling descent velocity |
| Fins fail to bond to the body tube | Ensure proper connection & build up epoxy |
| Fins misaligned for launch | Ensure proper alignment, connection, and epoxy build-up |
| Fins flutter during flight | Ensure proper alignment, connection, and epoxy build-up |
| Bulkhead disconnects/bursts | Ensure proper alignment, connection, and epoxy build-up |
| Centering ring(s) fail to bond with body tube | Ensure proper connection & build up epoxy |
| Eye bolt tears out of Centering ring | Ensure material is stable & has a strong bond |
| Retainer bursts | Handling the rocket carefully and controlling descent velocity |

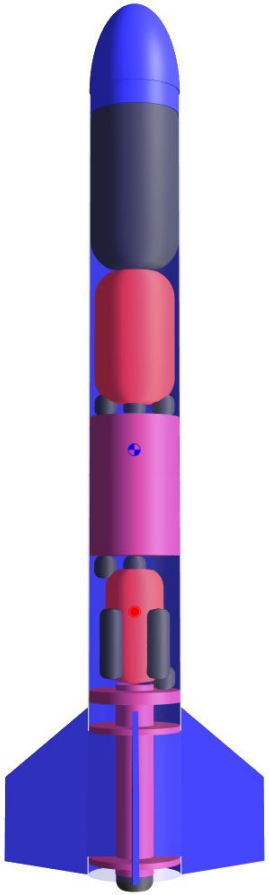
# Aerostructures

## Rocket Design

Rocket Materials:

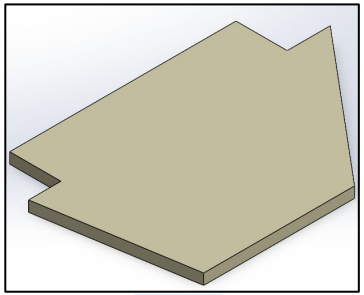
- Body Tube
  - Cardboard (4 in diam. | 34 in. L)
- Motor Tube
  - Cardboard ( 38mm diam. | 8 in. L)
- Nose Cone
  - PLA
- Fins, Centering Rings, Bulkheads
  - Birch Plywood (0.25 in thickness)
- Motor Mount
  - PLA



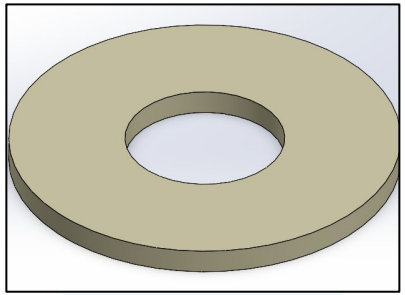


# Aerostructures

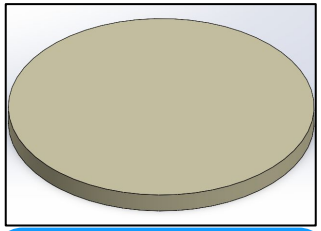
## Component Design



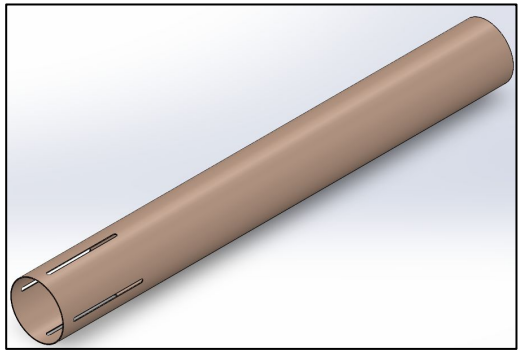
Fin(s)



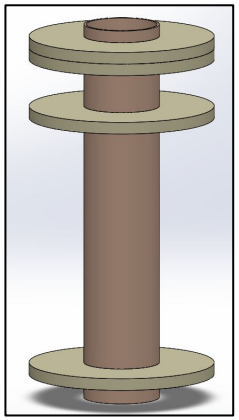
Centering Ring(s)



Bulkhead(s)



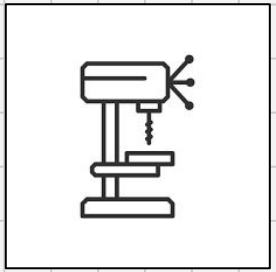
Body Tube w/ Fin Slots

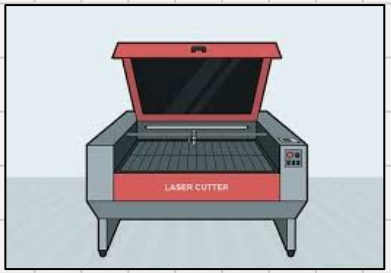


Motor Tube Assembly

# Aerostructures

## Manufacturing Plan



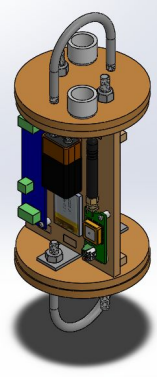


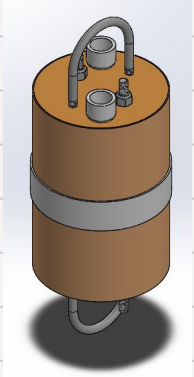

1. Fin slots will be drilled into the body tube using a drill press in the UCF machine shop.
2. Body tube will be split into 3 sections using a band saw in the UCF machine shop.
3. Fins, centering rings, and bulkheads will be laser cut in the TI lab.
4. Full build
    a. Motor tube structure with centering rings & u-bolt
    b. Fin attachment using epoxy adhesive
    c. Motor assembly to body tube
    d. Fix nose cone to top body tube third
    e. Assemble payload bays and insert into body tube



Example of Future State

# Avionics










## Requirements

| |
|---|
| The avionics bay shall weigh no more than 20 oz when fully configured. |
| The avionics system shall record flight data at a minimum frequency of 10 Hz for post-flight analysis. |
| The avionics system shall be capable of collecting data for a maximum of 5 minutes of flight. |
| The avionics system shall operate on a nominal voltage of 9 Volts. |
| The avionics system shall not exceed an amperage load of 500 mA at any point during operation. |
| The avionics system flight data shall be able to be extracted directly from the flight controller. |
| The avionics system shall include a backup recording device for flight characteristics. |
| The avionics bay shall integrate vent ports to accommodate onboard barometers. |

| |
|---|
| The avionics system shall employ a dual-deployment recovery setup. |
| The avionics system shall initiate the deployment of a recovery device at a minimum of 500 feet of altitude of flight descent. |
| The avionics system shall allow for the use of e-match pyrotechnic devices for recovery equipment initiation. |
| The avionics system should incorporate an onboard camera facing outward from the vehicle. |
| The avionics system should capture and store camera footage onboard the vehicle. |
| The avionics system onboard footage should be able to be extracted via microSD. |
| The avionics system shall be armed externally while the vehicle is on the launch pad. |
| The avionics system shall employ the use of an arming switch accessible through vent ports located on the side of the avionics bay. |
| Onboard electronics shall be properly connected, free of any burns or damages leading to the electronics being unreadable/unusable before or after flight. |

# Avionics

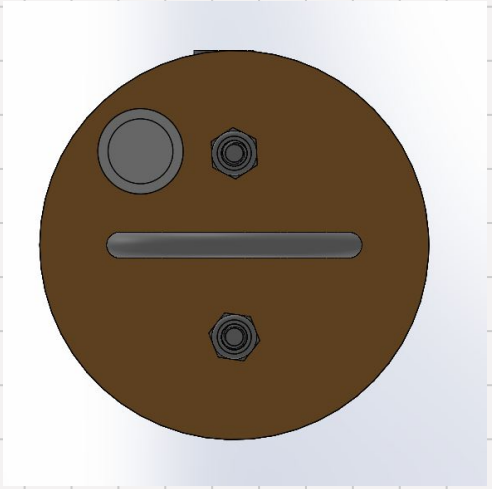
## Risks & Failure Modes

| Risks & Failure Modes | Mitigation Strategies |
|---|---|
| Damage to avionics bay integrity through vibration or tensile forces | Reinforced bay construction through using thicker wood and tight anchoring points |
| U-bolt shear out on bay splits halves of vehicle | Using a large u-bolt and base plate to distribute force from shock cord across bulkheads |
| Ejection charge ignition failure prevents proper separation | Redundant ejection charges to backup primaries |
| Premature avionics dropout due to power failure | Sizing batteries to proper milliAmp-Hour ratings for flight time |
| Wire deseating from terminal ports on altimeter and GPS | Tying down wires with zip ties to prevent vibration from deseating |
| Component mounting failure due to vibration during flight | Mounting to thicker boards to prevent flexion of the base |
| Unintentional switch toggle due to vibration during flight | Source switches intended or previously used in high power flight (rotary and COTS) |
| Humidity causing camera lens fogging and flight video quality loss | Ensure minimal temperature change and test footage before flight |

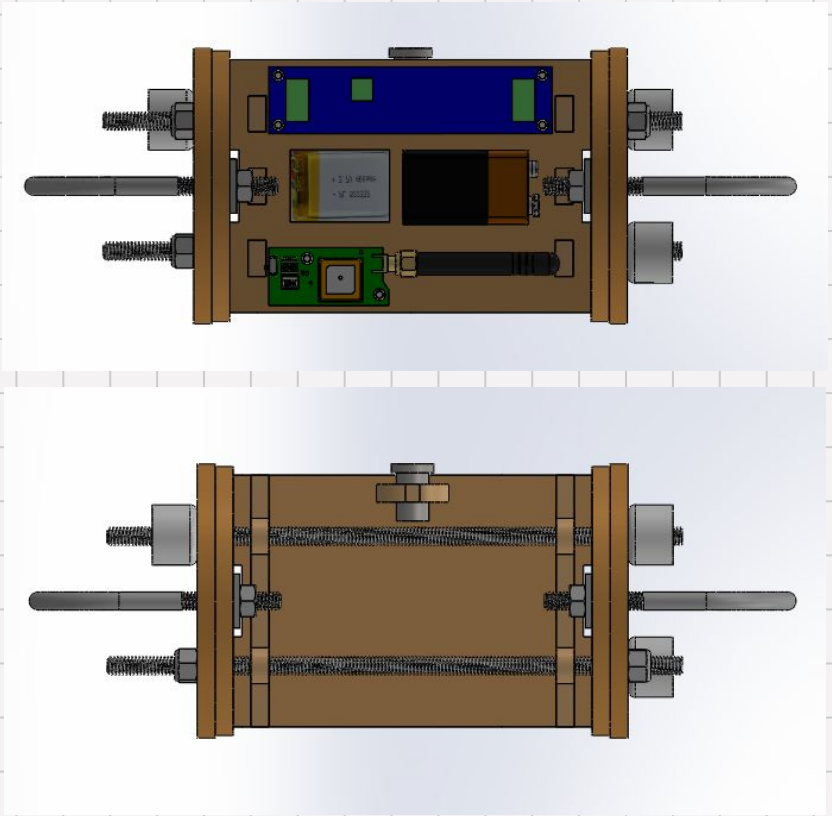
## Mitigation Strategies

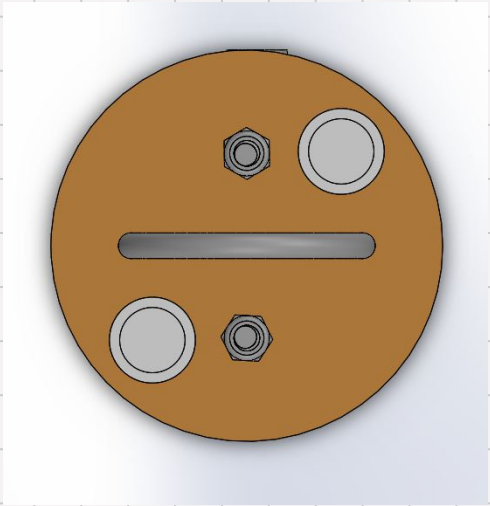
# Avionics

## Bay Design and Layout



Bottom





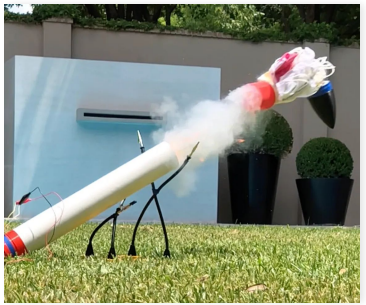
Top

# Avionics

## Manufacturing Plan

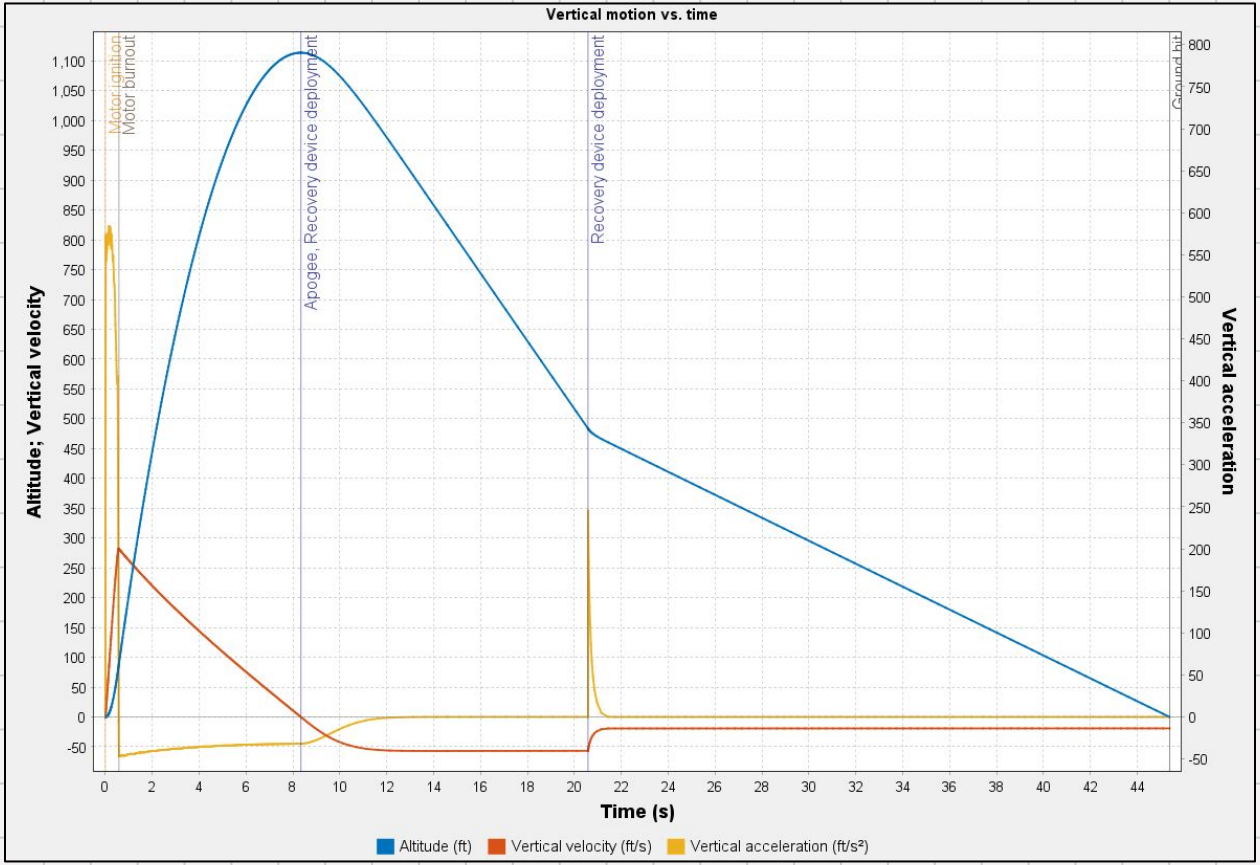
- Bench Testing Components
- Bay Manufacturing and Assembly
- Integrated Testing with Assembled Bay
- Vehicle Integration and Launch

Black Powder Ejection Testing

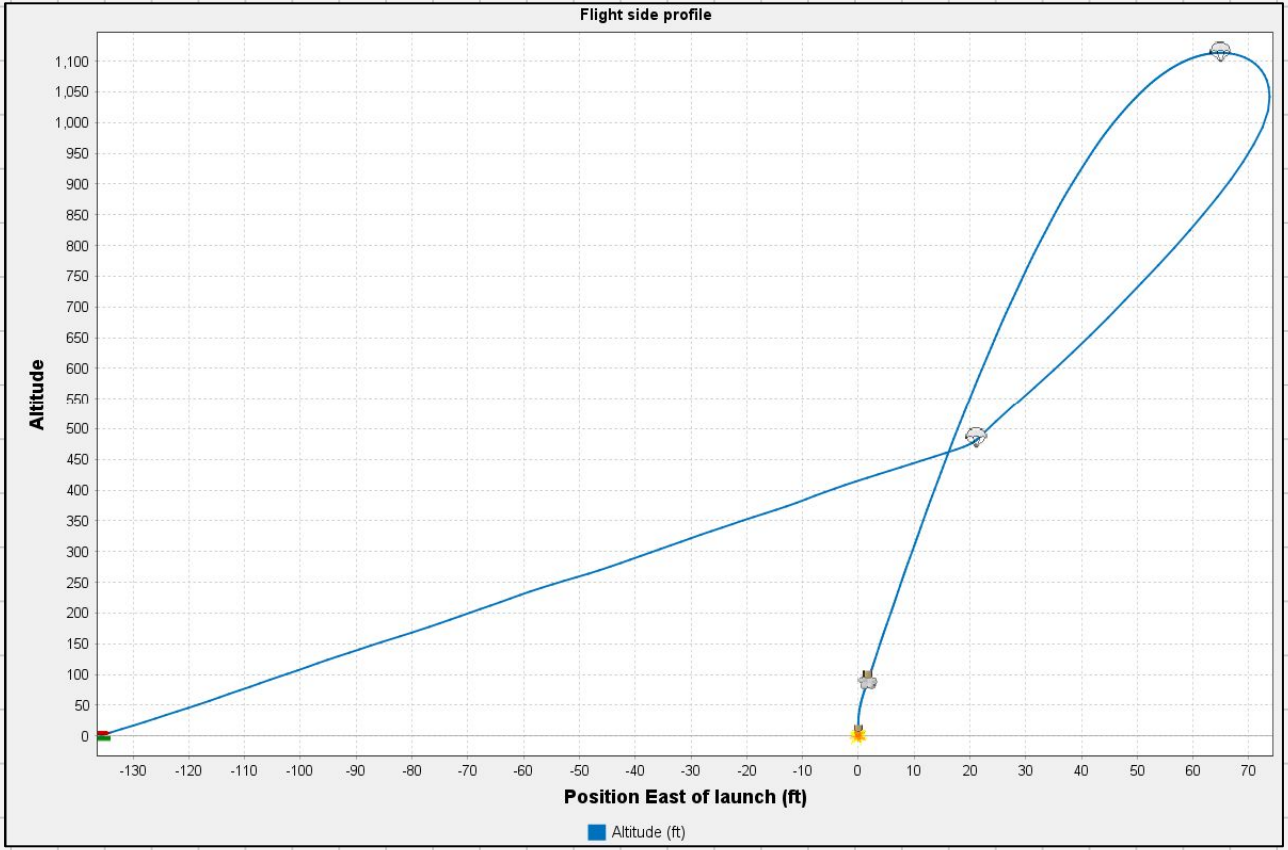


## Test?

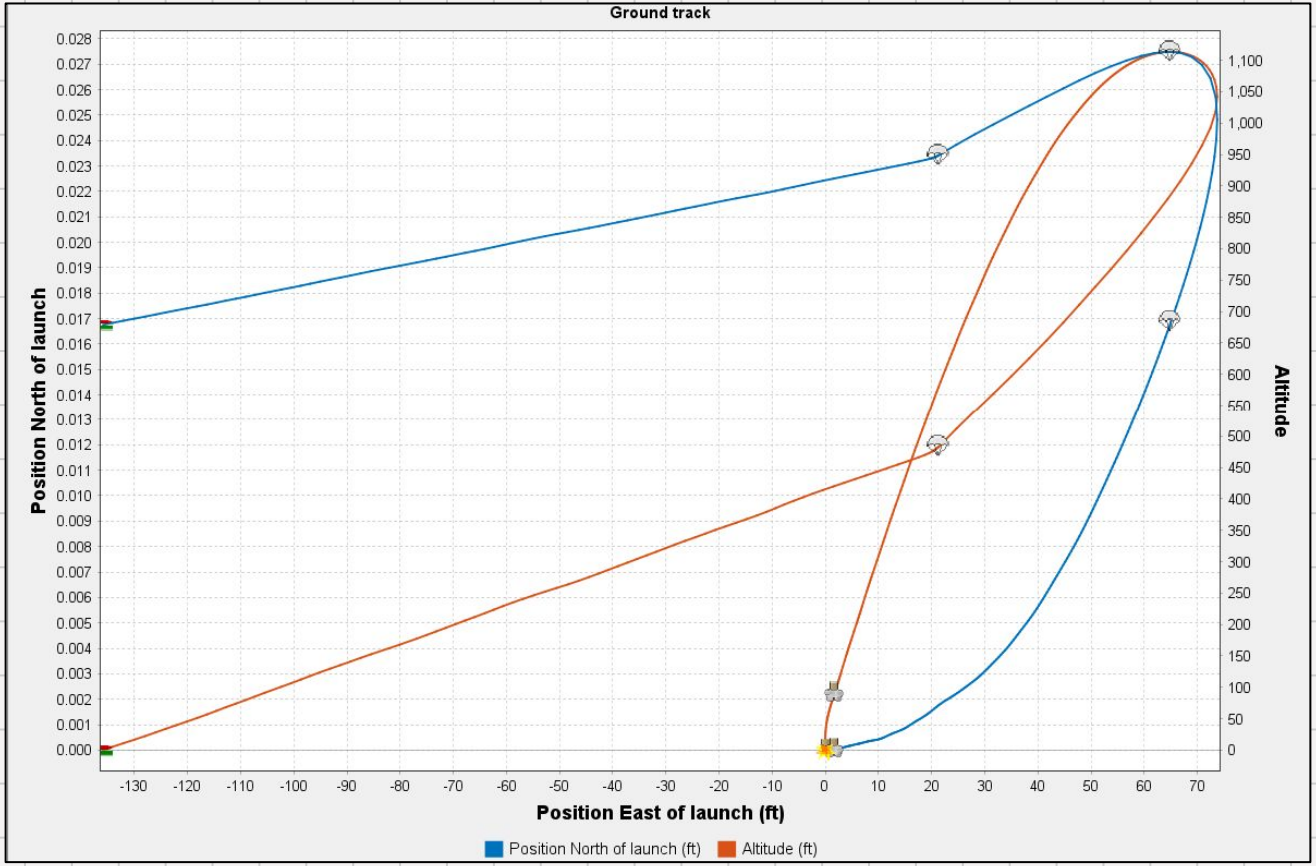
# Simulation - Vertical Motion vs. Time



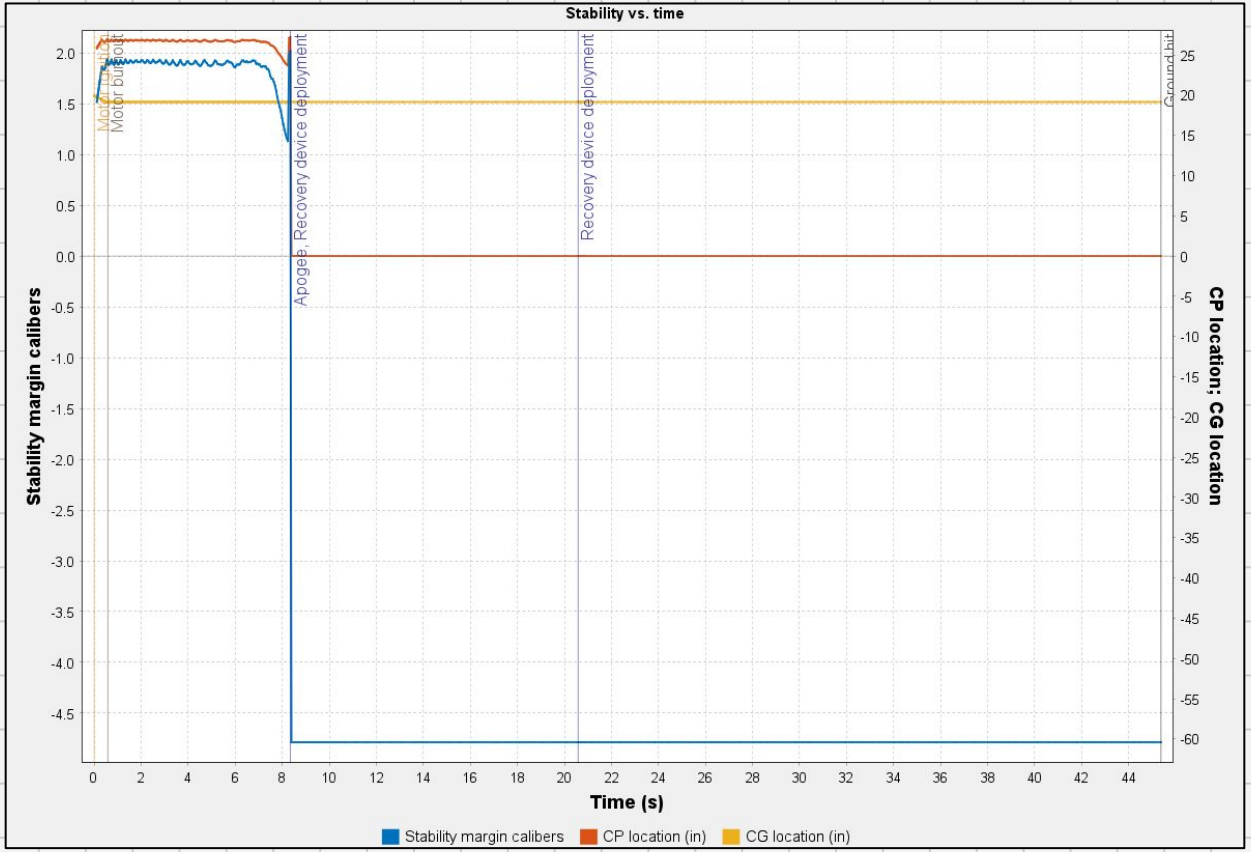
# Simulation - Flight Side Profile



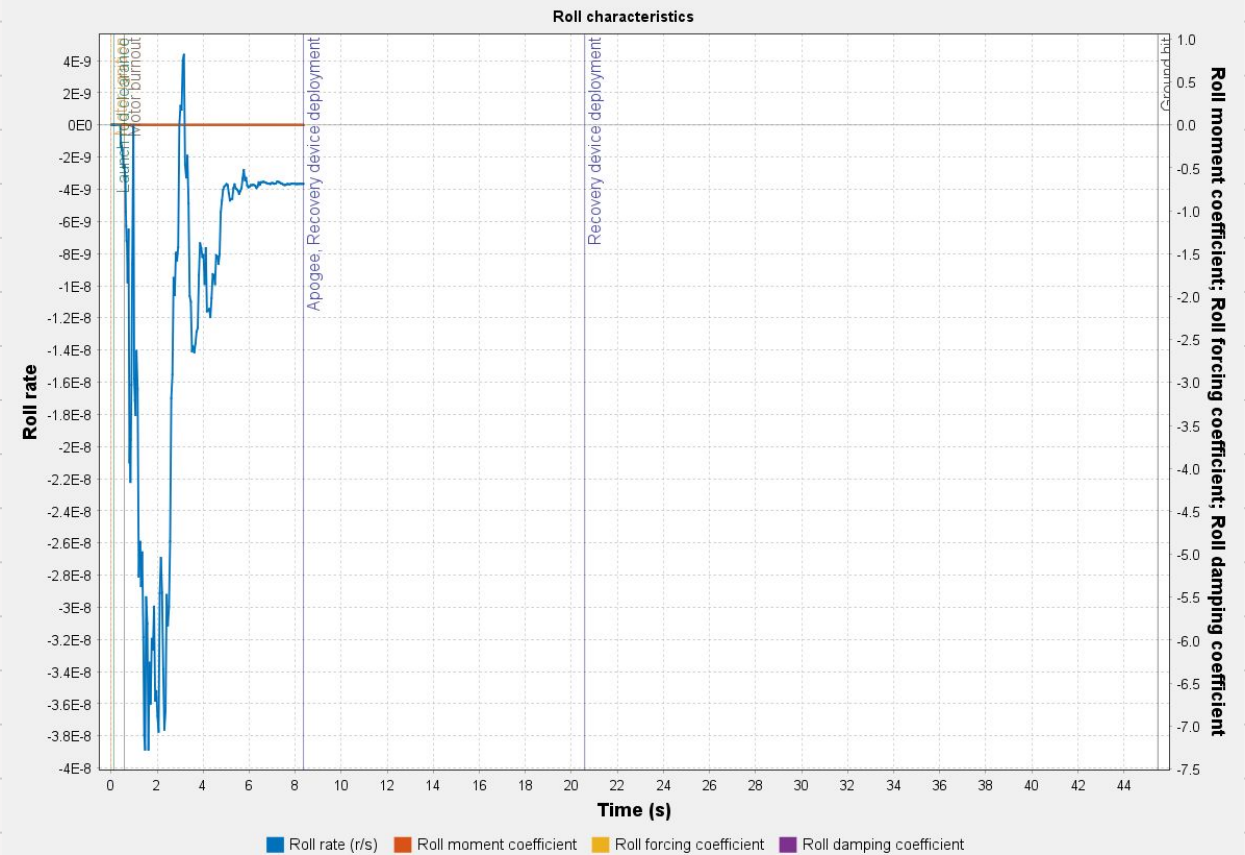
# Simulation - Ground Track



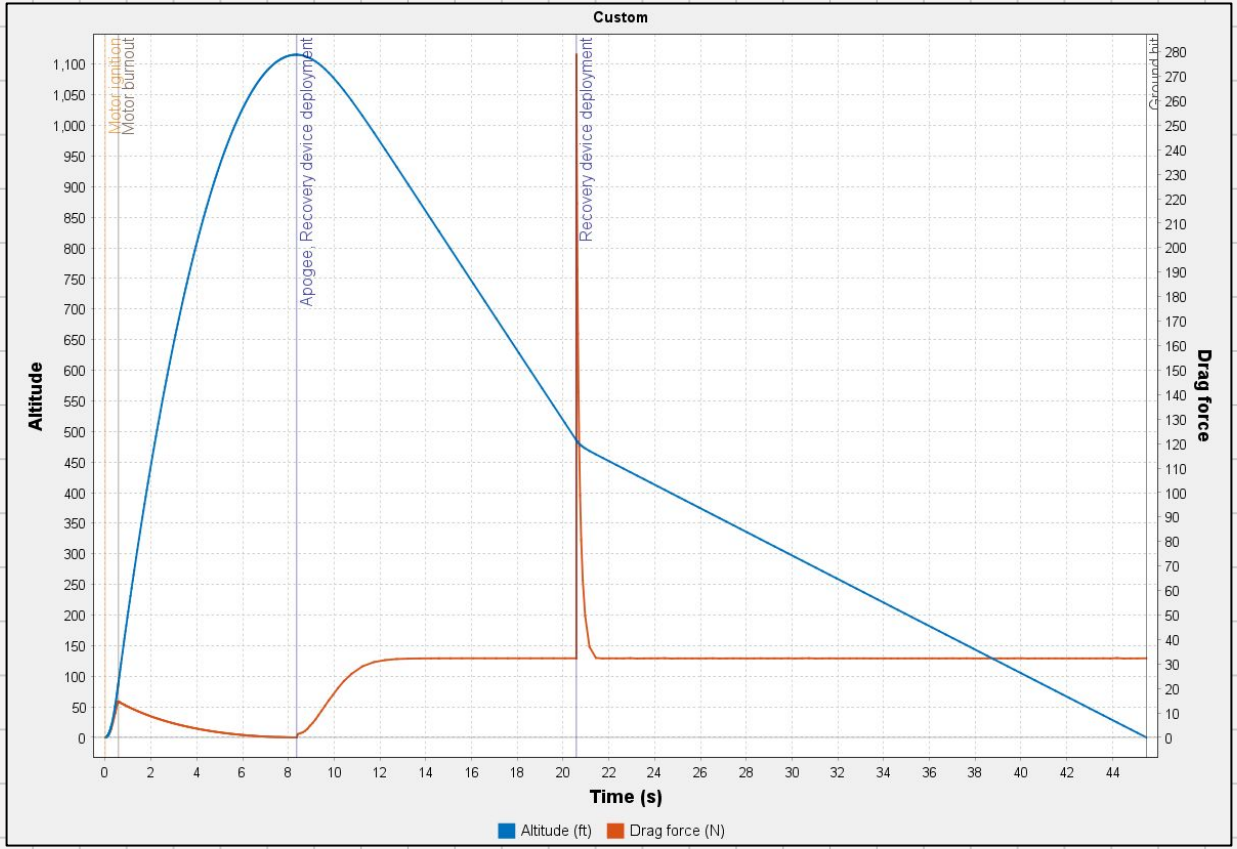
# Simulation - Stability vs. Time



# Simulation - Roll Rate



# Simulation - Drag Force During Flight



# Gantt Chart

| TASK | START | DAYS | END |
|---|---|---|---|
| **Manufacturing** | | | |
| Material and parts order | 1/12/2026 | 32 | 2/12/2026 |
| Material and parts acquisition | 2/12/2026 | 15 | 2/26/2026 |
| Body Tube Machining | 3/2/2026 | 2 | 3/3/2026 |
| Fin, Bulkhead, & Centering Ring Fabrication | 3/2/2026 | 1 | 3/2/2026 |
| Nose Cone Fabrication | 3/2/2026 | 3 | 3/4/2026 |
| Electronics Bay Fabrication & Assembly | 3/2/2026 | 5 | 3/6/2026 |
| Egg Payload Bay Fabrication | 2/24/2026 | 10 | 3/5/2026 |
| Water Payload Bay Fabrication | 3/2/2026 | 5 | 3/6/2026 |

# Gantt Chart

| TASK | START | DAYS | END |
|---|---|---|---|
| **Testing & Validation** | | | |
| Payload safety testing | 3/2/2026 | 12 | 3/13/2026 |
| Ejection Charge Testing | 3/2/2026 | 12 | 3/13/2026 |
| Shock Cord Load Testing | 3/2/2026 | 12 | 3/13/2026 |
| Payload Load/Unload Time Testing | 3/2/2026 | 12 | 3/13/2026 |
| Avionics Testing | 3/2/2026 | 12 | 3/13/2026 |
| Placeholder for last minute manufacturing tweaks | 3/13/2026 | 3 | 3/15/2026 |
| Placeholder for any additional testing needed | 3/13/2026 | 3 | 3/15/2026 |

# Gantt Chart

| TASK | START | DAYS | END |
|---|---|---|---|
| **Final Build** | | | |
| Project completion goal | 3/7/2026 | 15 | 3/21/2026 |
| Inner Tube & Centering Ring Fixing to Body Tube | 3/9/2026 | 5 | 3/13/2026 |
| Fin Fixing to Body Tube | 3/9/2026 | 5 | 3/13/2026 |
| Nose Cone Fixing to Body Tube | 3/9/2026 | 5 | 3/13/2026 |
| **Launch** | | | |
| Initial launch attempt | 3/21/2026 | 1 | 3/21/2026 |
| Final launch attempt | 4/18/2026 | 1 | 4/18/2026 |

# Gantt Chart

| TASK | START | DAYS | END |
|---|---|---|---|
| **Fallback Launch Attempt Plan** | | | |
| Additional Payload safety testing | 3/23/2026 | 19 | 4/10/2026 |
| Additional Ejection Charge Testing | 3/23/2026 | 19 | 4/10/2026 |
| Additional Shock Cord Load Testing | 3/23/2026 | 19 | 4/10/2026 |
| Additional Payload Load/Unload Time Testing | 3/23/2026 | 19 | 4/10/2026 |
| Additional Avionics Testing | 3/23/2026 | 19 | 4/10/2026 |
| Placeholder for last minute manufacturing tweaks | 4/10/2026 | 3 | 4/12/2026 |
| Placeholder for any additional testing needed | 4/10/2026 | 3 | 4/12/2026 |
| Final Build | 4/13/2026 | 5 | 4/17/2026 |

# Critical Path?

Team two is currently in the process of manufacturing and testing. With the goal of complete assembly by March 13.
Keeping pace to be ready by the competition on March 21!

(Assuming burn ban lifted)

